
BY 514: Analytical Biotechniques

Course Code	BY 514
Course Name	Analytical Biotechniques
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	B. Tech. 3rd and 4th year, MS/M. Tech., Ph.D.
Distribution	Core for M. Tech. Biotechnology, elective for others
Approval	9th Senate (May 2016)

Course Contents

- **Spectroscopy**

Principles and applications of UV-Visible spectroscopy, circular dichroism, fluorescence spectroscopy, mass, and infrared spectroscopy, MALDI-TOF, NMR. [12 Lectures]

- **Chromatographic and other separation techniques**

Principles and applications of different chromatographic techniques, ultrafiltration, phase-partitioning, Gel electrophoresis, two dimensional gel electrophoresis, blotting techniques. [10 Lectures]

- **Imaging Techniques**

Principles and applications of bright-field, dark-field and phase contrast microscopy, fluorescence microscopy, confocal microscopy, electron microscopy and atomic force microscopy. [12 Lectures]

- **Module IV**

Principles and applications of Surface plasmon resonance, Flow cytometry, Real Time PCR, ELISA. [8 Lectures]

Text and Reference Books

1. Charles R. Cantor and Paul Schimmel, **Biophysical Chemistry, Part 2: Techniques for the Study of Biological Structure and Function (Pt. 2)**.
2. Lakowicz, Joseph R. **Principles of Fluorescence Spectroscopy**, 3rd Edition.
3. Michael Hoppert, **Microscopic Techniques in Biotechnology**, 2006.
4. K. Wilson and J. Walker, **Principles and Techniques of Biochemistry and Molecular Biology**